

Data Analytics Capstone

Air Quality Risk Prediction Using Environmental and Meteorological Data

Final Report

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The GitHub repository for this capstone project is available at:

<https://github.com/KlariVZ/air-quality-risk-prediction>

The repository includes the raw data files, processed datasets, notebooks, figures, tables, and supporting project documentation used for the Interim Report.

Air Quality Risk Prediction Using Environmental and Meteorological Data

Abstract

Poor air quality is a particularly challenging outcome to classify due to the complex interplay of a variety of factors occurring temporally, seasonally, meteorologically, and in terms of pollution. This study aimed to develop a framework for same-day binary classification to identify poor air quality days, evaluate predictors that play a key role, examine variation in seasons, and ultimately examine the performance of various machine learning models side by side. In order to develop this framework, days classified with elevated poor air quality in Kapolei, Honolulu County, Hawaii (hereafter referred to as the study site) were investigated at a local strictly level, utilizing environmental and weather data that is available to the public.

The U.S. Environmental Protection Agency AirData (EPA) / Air Quality System's (AQS) daily air quality data and the National Oceanic and Atmospheric Administration (NOAA) Climate Data Online's (CDO) daily meteorological summaries served as the source of the data for this study, and spanned January 1, 2024, to December 31, 2025 (hereafter referred to as the study period). The finalized dataset was created thereafter, and consisted of the following:

- 731 daily observations that covered the study period.
- Each presented observation as represented by one calendar date.
- Pollutant variables:
 - Ozone

- PM2.5
- PM10
- Nitrogen Dioxide (NO₂)
- Meteorological variables:
 - Maximum temperature
 - Minimum temperature
 - Precipitation
 - Average wind speed

For the study, a threshold specific to the study in relation to locally elevated AQI conditions, was used, namely; ``overall_aqi > 35``. From there ``poor_air_quality``, the binary target variable, was derived from ``overall_aqi``.

Data cleaning, exploratory data analysis (EDA), feature engineering, model development, and model evaluation was included in the analytical workflow. In order to clean the data, prepare the dataset for modeling, train the models, visualization generation, and reproducible documentation outputs, the following facilitated data preparation:

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Imbalanced-learn
- Google Colab

- Github

Four classification models were chosen and evaluated: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM). The class-weighted Random Forest model outperformed the other classifiers with an accuracy of 0.986, precision of 0.967, F1-score of 0.967, and ROC-AUC of 0.994 on the holdout test set. Experimentation also revealed that Synthetic Minority Oversampling Technique (SMOTE) as well as hyperparameter tuning did not improve or enhance the preferred model. Chronological validation resulted in an F1-score of 0.947, and an ROC-AUC of 0.996, which in turn confirmed model stability. A significant seasonal variation in poor air quality was revealed by inferential testing, but yielded no significant contrast for weekday or weekend status. Potentially, the framework may facilitate local environmental observation, eventual decision support interfaces, and sustainability blueprinting.

Introduction

Background and Context

The climate and sustainability domain continues to grapple with the prevailing issue of poor air quality because of its effects on public health, decisions for local planning, and the conditions of the environment. Particularly in the case of fine particulate matter such as PM_{2.5}, air pollution has been linked to various cardiovascular and respiratory health concerns in recent literature, and a continued emphasis is placed on the importance of effective surveillance and prediction of air quality (Wan Mahiyuddin et al., 2023). More recently, as the availability of environmental data has improved, research in air quality has

seen an increase in direction towards predictive and decision facilitating approaches in order to more proactively identify conditions of elevated risk (Wang et al., 2025).

A multitude of interacting environmental and meteorological variables influence the state of air quality, including pollutant densities, rainfall, temperature, seasonal circumstances, wind velocity, and temporal formations. It is a given that these factors will vary across time and location, therefore, if a structured analytical framework is not in place it becomes extremely difficult to accurately anticipate the conditions of poor air quality with consistency. Weather conditions, which is an amalgamation of meteorological variables and seasonal circumstances, can impact the ways in which pollution accumulates, disperses, and concentrates. As a result, the use of data containing meteorological variables and seasonal fluctuation, is imperative when attempting to predict air quality and is substantiated by prior research (Eren et al., 2025).

The unit of analysis for the study site for the duration of the study period, is observation made at the daily level, with every individual row representing a single calendar date. The study uses a combination of raw air quality and meteorological data collected from the public EPA and NOAA databases. The study does not execute future day forecasting. A same day binary classification framework is developed in its stead that classifies whether a particular day presents with poor air quality that is elevated locally deduced from weather, pollutant, and temporal variables.

Practically, this study adds value in its translation of numerous pollutant and meteorological variables into a simpler classification outcome. Environmental monitoring teams, sustainability planners, local government staff, and public information units may gain a better understanding of which daily conditions inform elevated air quality risk with this

type of framework. However, it should be clarified that the framework is not a substitute for official air quality reporting or regulatory systems, but is proposed as an aid for observation and decision guidance.

Problem Statement

The objective of this study is to classify $Y = \text{'poor_air_quality'}$, a binary variable derived from 'overall_aqi' , for daily observations in Kapolei, Honolulu County, Hawaii (the study site), using $X =$ pollutant concentration variables, meteorological variables, and temporal features over the period January 1, 2024, through December 31, 2025 (the study period). Success is evaluated using classification metrics including accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, cross-validation, and sensitivity checks for class imbalance and time based validation.

Purpose of the Study

The purpose of this study is to develop and evaluate a predictive classification framework for same day identification of days with locally elevated poor air quality through the use of environmental and meteorological data. More precisely, this study intends to identify the pollutant, weather and temporal variables most closely linked to days of poor air quality; assess if it is possible to correctly classify those days using the predictors that were selected; scrutinize whether days of poor air quality differ substantially across seasons and time periods; and determine which model performs most optimally by the comparison of candidate machine learning models.

Research Problems / Research Questions

To guide the study, the following research questions are posed:

- Research Question 1 (RQ1): Which environmental and meteorological variables are most strongly associated with poor air quality days?
- Research Question 2 (RQ2): Can poor air quality days be accurately identified and classified using pollutant and weather data?
- Research Question 3 (RQ3): Do poor air quality days vary significantly across seasons and time periods?
- Research Question 4 (RQ4): Which predictive model performs best for the binary classification of poor air quality?

Contributions and Expected Value

Practical contribution: This study offers a feasible roadmap, using accessible EPA and NOAA data, for classifying days of poor air quality locally. The more efficient interpretation of daily pollutant and weather conditions from the classification output may prove valuable to teams that monitor the environment as well as sustainability stakeholders.

Technical contribution (methods/models): This study contributes a replicable data analytics workflow that incorporates data cleaning, feature engineering, EDA, inferential testing, machine learning model comparison, class imbalance evaluation through the use of SMOTE, hyperparameter tuning, and chronological validation. The candidate models include Logistic Regression, Decision Tree, Random Forest, and SVM. Rather than using accuracy alone, multiple classification metrics were used for the definitive model selection.

Value to stakeholders/target users: The value expected as an outcome of this study is the enhanced constructability of localized patterns in air quality and stronger advancement for tracking and planning direction. Users that could benefit from a digestible classification framework that pinpoints days with localized elevated air quality risk and emphasizes the variables most pertinent to that classification.

Literature Review

Literature Review Approach

The literature that was deemed pertinent to the study's focus on air quality classification, pollutant and meteorological predictors, seasonal variation, class imbalance cognizant model examination, and machine learning model evaluation were chosen on the basis that they satisfy at least one of the four research questions. Combinations such as "air quality prediction", "prediction of AQI", "PM2.5 prediction", "meteorological variables", "air pollution concentrations across seasons", "random forest model air pollution", "support vector machine AQI", "machine learning air quality", and "air pollution health impacts". Peer reviewed studies that have publications in journals related to machine learning, public health, environmental science, and atmospheric science were given precedence.

Research focused on air pollution monitoring, health impact of pollutant exposure, meteorological influences, variation based on season or time, machine learning model comparison, feature importance, or explainability all informed the inclusion criteria for the literature review. Furthermore, the curated literature substantiated the data sources, predictor selection, classification approach, model choices, and evaluation strategy for the study. Albeit the use of language in some reviewed studies were forecasting centered, this study is

designed as a same day binary classification study as opposed to a future day forecasting model. It is important to highlight this distinction as the current methodology classifies daily poor air quality status using same day pollutant, weather and temporal predictors and not lagged or forecasted inputs to predict future AQI values.

Summary of Key Literature

Wan Mahiyuddin et al. (2023) reviewed the association of fine particulate matter, in particular PM_{2.5}, with cardiovascular and respiratory health effects with the purpose of synthesizing evidence derived from time-series studies on PM_{2.5} and the resulting health outcomes. The review supports the pragmatic significance of identifying days with elevated air quality concern, which makes the article relevant to this study. Although health outcomes are not modeled by this study, the relevance of broader environmental and public health is supported by the literature.

Islam et al. (2025) investigated air pollution and cardiovascular disease, also addressing exposure from sources such as bushfire smoke, and examined their relationship. The approach of this study was a systematic review and highlighted that air pollution is associated with major concerns relating to health. Therefore it supports the feasible motivation and background for this project. Although this project's focus is on environmental data analytics rather than modeling for direct health risk, its significance is the most compelling in its explanation of why air quality classification may matter to stakeholders.

Eren et al. (2025) studied seasonal analysis and prediction of air pollutants in relation to meteorological parameters using machine learning based methods. Therefore, it is closely aligned with this project because of the connection it makes between air pollutants, seasonal

fluctuation, weather variables and machine learning. The inclusion of meteorological variables and seasonal analysis is supported by the findings of this study. It is particularly applicable to whether poor air quality days have significant variation across seasons and time, which is examined by RQ3.

Kamińska (2018) took traffic and meteorological conditions then employed Random Forest methods to short term air pollution modeling, which is relevant because the application of tree based ensemble methods was demonstrated in air pollution analysis and the significance of feature importance during interpretation. Random Forest as a candidate model for this project and the use of feature importance results for RQ1, is therefore supported by this study. It should be noted that the study also points to model results being tied to a specific location as a result of local traffic, geography, and meteorology affecting pollutant behavior.

Qi et al. (2024) approached meteorological normalization of PM_{2.5} by comparing linear regression, generalized additive models, and Random Forest methods. The demonstration of how different statistical and machine learning methods, subject to the modeling objective and data structure, can generate varied results makes this source important in its support of multiple model comparison instead of relying on a singular method.

Sachdeva et al. (2024) constructed an AQI prediction approach with the use of pollutant concentration and meteorological variables, which is in support of pollutant and weather variable integration in this project. Also, because it supports the concept that environmental and meteorological predictors can be used in modeling to produce AQI related

outcomes, it is relevant to RQ2. Nevertheless, it also mirrors a frequent air quality modeling obstacle:

Sigamani and Venkatesan (2022) investigated AQI prediction in the context of Internet of Things (IoT) applications by making use of meteorological parameters and machine learning. The study bears technical relevance to the execution side of this project as it bridges air quality modeling with monitoring systems and strategic planning frameworks, which feasibly validates the proposal of a future operational dashboard or monitoring support system. By the same token, it is underscored by the study that practical deployment requires critical attention to the data availability, system design, and robust evaluation that extends beyond the independent assessment of model accuracy alone.

Tırink (2025) comparatively examined machine learning methods, including SVMs, for AQI forecasting under long term environmental dynamics. Therefore, it aligns with this project's comparative scope, provides a theoretical anchor for the inclusion of SVM as a candidate classifier, as well as emphasizes the importance of utilizing multiple evaluation metrics. For this project, accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, cross-validation, Smote, and chronological validation were all used to ensure more balance during assessment.

Wang et al. (2025) discussed temporal data driven PM2.5 forecasting and bridged theoretical forecasting frameworks with empirical analysis, which makes it relevant to this project by placing specific emphasis on the necessity of temporal structure in air quality modeling. At this juncture, it also points to an important distinction for this current project, as this project uses daily observations and temporal features, it does not purport to operate as a true forecasting model. Instead, this project is positioned as a same-day classification study.

In future, a design which incorporates lagged variables, rolling variables, and future weather inputs could extend its functionality towards forecasting.

Zhou et al. (2024) reviewed deep learning paradigms for PM2.5 concentration prediction, noting the profound complexity of pollution modelling. The study demonstrates that advanced models can effectively be applied to air quality challenges, yet at the same time raises concerns related to model complexity, interpretability, and evaluation, which makes this source very useful. It should be noted that deep learning was not used for this capstone. The final dataset for this project contained 731 daily observations, which makes it unsuitable for deep learning, which is in support of the project's decision in prioritizing more interpretable traditional machine learning models such as Logistic Regression, Decision Tree, Random Forest, and SVM.

Table 1

Literature Relevance Matrix

Author(s)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	Support for RQ(s) / Project Decisions
Wan Mahiyuddin et al. (2023)	PM2.5 and health effects	Time-series health studies	Literature review	PM2.5 is linked to cardiovascular and	Supports the practical importance of identifying

				respiratory health concerns	elevated air quality days
Islam et al. (2025)	Air pollution and cardiovascular disease	Public-health literature	Systematic review	Air pollution is associated with major cardiovascular concerns	Supports the background and stakeholder relevance of air-quality monitoring
Eren et al. (2025)	Air pollutants, meteorology, and seasonality	Sakarya, Türkiye	Seasonal analysis and machine learning	Meteorological parameters and seasonal patterns are relevant to pollutant prediction	Supports RQ3 and inclusion of weather and seasonal variables
Kamińska (2018)	Short-term air pollution modeling	Wrocław case study	Random Forest	Random Forest can model air	Supports Random Forest and

				pollution patterns and provide feature importance	feature-importance interpretation for RQ1 and RQ4
Qi et al. (2024)	PM2.5 meteorological normalization	PM2.5 and meteorological data	Regression, GAM, Random Forest	Model results vary by method, supporting comparative modeling	Supports model comparison and interpretability
Sachdeva et al. (2024)	AQI prediction	Pollutant and meteorological variables	AQI prediction model	Pollutant and weather variables can support AQI-related prediction	Supports RQ2 and merging EPA and NOAA variables
Sigamani & Venkatesan (2022)	AQI prediction for monitoring systems	IoT air-quality context	Machine learning	AQI prediction can support monitoring	Supports implementation and

				and decision-support systems	dashboard potential
Tırink (2025)	Long-term AQI forecasting	Environmental time-pattern data	XGBoost, LightGBM, SVM	Comparative ML evaluation supports model selection	Supports RQ4 and inclusion of SVM and multiple metrics
Wang et al. (2025)	PM2.5 time-series forecasting	PM2.5 time-series data	Data-driven forecasting framework	Temporal structure is important in air-quality modeling	Supports use of time features and future forecasting extensions
Zhou et al. (2024)	PM2.5 prediction complexity	Deep-learning literature	Review of deep-learning architectures	Advanced models can be powerful but may reduce interpretability	Supports avoiding deep learning for this small, interpretable

					capstone dataset
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The reviewed literature, as well as validating the necessity of air quality analysis, also demonstrated the methodological complexity of air quality modeling. Identifying elevated air quality conditions matter as demonstrated by studies focused on health, especially when pollutants such as PM_{2.5} are present. However, health outcomes or casual effects are not inferred by this capstone. Concomitantly, the researched health literally plays an important role in justifying why monitoring and classifying locally elevated air quality days are of practical relevance.

A second theme highlights the necessity of using pollutant and meteorological variables in combination and is supported by several studies highlighting certain weather variables in air quality modeling such as precipitation, wind velocity, temperature, and seasonal signifiers. For this study, because the final dataset combines EPA pollutant records and NOAA meteorological summaries, this second theme is very relevant. That being said, it is also suggested by the literature that differences by location, pollutant, and time period can be seen in the effect of meteorological variables. Hence, it should not be assumed that model findings can be generalized from one geographic setting.

A third theme is machine learning model comparison having both value and limitation. As shown by previous studies, for AQI or pollutant modeling, Random Forest, SVM, regression models, and approaches that are more advanced can all be used. This is aligned with this study's decision to compare Logistic Regression, Decision Tree, Random

Forest, and SVM. By the same token, careful scrutinization of model performance is advised by the literature. Using accuracy as the overarching metric of success is not sufficient, as high accuracy may be a result of local data structure, predictor-target overlap, and instead of general operational readiness, a reflection of class distribution.

A fourth theme is predictive performance and interpretability having a tradeoff. It is shown in deep learning studies that advanced architectures used for air quality prediction tasks may improve performance, especially if the datasets are larger and more complex. Conversely, transparency for these models can fall short and the larger datasets required for these models are not available in this study. Interpretable and feasible models are more appropriate for a daily dataset that contains 731 observations. Random Forest, in particular, proves especially useful because of robust predictive performance, and availability of feature importance measures.

Lastly, the need to clearly distinguish classification from forecasting is reinforced by the literature as many studies pertaining to AQI use forecasting language as they predict future AQI values and pollutants. Lagged predictors or future weather forecasts are not used in this capstone, thus it should not be interpreted as a forecasting tool. This study is a framework for same day binary classification, but future research could add lagged pollutant measures, rolling averages, future meteorological forecasts, and chronological validation across additional years could extend it into forecasting.

Materials and Method

Study Design

This study followed a quantitative, observational, daily level data analytics design guided by an analytical objective formulated as a same-day binary classification, and not future day forecasting. In the final dataset each row was representative of one calendar date for Kapolei, Honolulu County, Hawaii (study site), spanning the period of January 1, 2024 through December 31, 2025 (study period). For this study, a combination of EDA, inferential testing, class imbalance evaluation, hyperparameter tuning, chronological validation, and supervised machine learning model comparison.

The model utilized `poor\_air\_quality` as the dependent variable, and `overall\_aqi` served as the binary target variable. Other variables including pollution concentration measures, meteorological variables, and temporal features were independent. The model's primary objective, given a daily observation, was to classify whether it represented a locally elevated poor air quality day.

Data Sources

Air quality data were obtained from the U.S. Environmental Protection Agency AirData (EPA) / Air Quality System (AQS) daily files. Provided in these files were daily pollutant concentration values and pollutant specific AQI values for ozone, PM2.5, PM10, and nitrogen dioxide (NO2). The meteorological data were obtained from the National Oceanic and Atmospheric Administration (NOAA) Climate Data Online's (CDO) daily meteorological summaries. Provided within the NOAA file was maximum temperature, minimum temperature, precipitation, and average wind speed.

The final cleaned dataset contained 731 daily observations. The EPA monitoring site was Kapolei, Honolulu County, Hawaii (study site). For meteorological variables, the Honolulu International Airport, HI US, NOAA weather station was used, which provided viable daily weather observations for the study period. The data files, cleaned datasets, notebooks, figures, and output tables are available in the GitHub repository listed after the title page.

Inclusion and Exclusion Criteria

The study included observations between January 1, 2024 through December 31, 2025 (study period). EPA pollutant files were included if they contained relevant daily observations for the Kapolei monitoring site as well as one of the pollutants: ozone, PM2.5, PM10, or NO2. NOAA observations were included if they provided daily meteorological variables applicable to the nearby weather station.

Exclusion of data included records outside of the study period. For analysis, pollutants not included in the selected EPA daily files were excluded. Any removal of rows was not done solely as a result of a specific pollutant or weather being missing; instead, during the modeling stage missing numeric predictor values were handled using median imputation.

Data Cleaning and Preparation

The raw EPA and NOAA files were cleaned in Python using Google Colab. Date fields were standardized into a common `date` variable in order for all data sources to be merged successfully at the daily level. Relevant variables were retained, namely: pollutant, AQI, observation-count, percent-complete, site metadata, and weather.

Some dates had duplicate PM2.5 records as a result of multiple parameter occurrence code values. Daily records were aggregated seeing as one row per date served as the unit for analysis. PM2.5 concentration values were averaged by date, while the PM2.5 AQI value was summarized using the daily maximum daily AQI value. The daily structure was therefore preserved whilst retaining the higher AQI contribution for target construction.

To preserve the full study period, a complete daily calendar from January 1, 2024 through December 31, 2025 was used. Date was used to merge the cleaned EPA pollutant files and the NOAA weather file. Duplicate dates, missing values, date range, and target consistency was used to check the merged dataset.

Feature Engineering and Target Construction

The variable `overall\_aqi` was calculated, for each date, as the maximum available pollutant specific AQI value across ozone, PM10, PM2.5, and NO2. The binary target variable `poor\_air\_quality` was then derived from `overall\_aqi`. The final target rule was:

- `poor\_air\_quality` = `when overall\_aqi > 35; otherwise poor\_air\_quality 0`

This threshold is a study specific operational definition of locally elevated AQI conditions, and is not to be interpreted as an official EPA health category cutoff. After a sensitivity review, the final threshold was selected. Only 12 poor air quality days were identified by the threshold of `overall\_aqi > 50`, only 1.65% of the dataset, which as a result created a severe class imbalance and limited the value of model training. After review and selection, 150 poor air quality days were identified by the threshold of `overall\_aqi > 35`, or 20.52% of the dataset, while still being representative of locally elevated AQI conditions within the study period for Kapolei.

From the `date` variable, additional temporal features were created: `year`, `month`, `day\_of\_week`, `is\_weekend`, and `season`. Seasons were assigned with the use of calendar months: winter included December, January, and February; spring included March, April, and May; summer included June, July and August; and fall included September, October, and November.

Predictor and Target Variables

Table 2

Modeling Target

Variable	Role	Description
poor_air_quality	Target	Binary variable where 1 indicates overall_aqi > 35 and 0 indicates overall_aqi ≤ 35

Table 3

Final Modeling Predictors

Predictor	Type	Source
daily_max_8hr_ozone	Pollutant concentration	EPA
daily_mean_pm10	Pollutant concentration	EPA
daily_mean_pm25	Pollutant concentration	EPA

daily_max_1hr_no2	Pollutant concentration	EPA
tmax	Meteorological	NOAA
tmin	Meteorological	NOAA
prcp	Meteorological	NOAA
awnd	Meteorological	NOAA
month	Time-based feature	Derived
is_weekend	Time-based feature	Derived

Because of their use in the target construction, pollutant specific AQI variables and `overall\_aqi` were excluded from the predictor set, which reduced direct target leakage. However, conceptually pollutant concentration predictors remain related to AQI construction and are addressed as a limitation in the methodology.

Missing Value Treatment

After merging the EPA and NOAA files, missing values were reviewed. For data availability to be accurately described, missing values were retained. For modeling, median imputation was used for missing numeric values. Median is transparent, easily reproducible, and has less sensitivity towards extreme values than mean imputation, and was chosen based on this criteria. As a result, modeling and comparison had 731 daily observations available.

Research Questions and Hypothesis

Four research questions guided this study and hypotheses were evaluated at the 5% significance level.

Table 4

Research Questions and Hypothesis

Research Question	Null Hypothesis	Alternative Hypothesis	Primary Method
RQ1: Which environmental and meteorological variables are most strongly associated with poor air quality days?	H ₀₁ : Environmental and meteorological variables are not meaningfully associated with poor air quality days.	H _{a1} : At least one environmental or meteorological variable is meaningfully associated with poor air quality days.	Correlation analysis and Random Forest feature importance
RQ2: Can poor air quality days be accurately identified and classified using pollutant and weather data?	H ₀₂ : Poor air quality days cannot be classified better than a baseline or chance-level approach.	H _{a2} : Poor air quality days can be classified using pollutant, weather, and time-based variables.	Supervised binary classification

RQ3: Do poor air quality days vary significantly across seasons and time periods?	H ₀₃ : Poor air quality conditions do not differ significantly across seasons or time periods.	H _{a3} : Poor air quality conditions differ significantly across seasons or time periods.	Kruskal-Wallis, Mann-Whitney U, and chi-square tests
RQ4: Which predictive model performs best for the binary classification of poor air quality?	H ₀₄ : There is no meaningful performance difference among the candidate classification models.	H _{a4} : At least one model performs better than the others.	Model comparison using classification metrics

Sample Size Justification

Separate estimation was done for each research question to estimate the minimum sample size using power-based or confidence-level-interval-based logic. The largest minimum requirement was selected as the overall minimum study size.

Table 5*Minimum Sample Size Computation by Research Question*

Research Question	Method Used	Key Parameters	Minimum Sample Size
RQ1	Green's rule for multiple regression	$50 + 8m$, with $m = 8$ predictors	114
RQ2	Confidence interval for a proportion	$Z = 1.96$, $e = 0.05$, $p = 0.50$	385
RQ3	Power estimate for one-way ANOVA	$\alpha = 0.05$, power = 0.80, effect size $f = 0.25$, groups = 4	180
RQ4	Confidence interval for expected model accuracy	$Z = 1.96$, $e = 0.05$, expected accuracy = 0.80	246

The largest minimum sample size was 385 observations and with the final cleaned dataset containing 731 daily observations, it exceeded the minimum requirement. Descriptive analysis, inferential testing, train-test splitting, cross-validation, and model comparison was supported by this sample size.

Exploratory Data Analysis

To gain an understanding of the data before modeling, EDA was conducted and included descriptive statistics, target distribution review, missing value review, seasonal summaries, monthly summaries, dominant pollutant analysis, correlation analysis, and visualizations of AQI trends over time.

Methodological decisions were also informed by EDA such as the threshold sensitivity review that revealed that the official AQI-related threshold of ``overall_aqi > 50`` produced too few positive cases for stable classification. The inferential testing approach for RQ3 was also supported by EDA revealing seasonal variation in AQI patterns.

Inferential Testing for RQ3

RQ3 was evaluated using inferential tests at $\alpha = 0.05$. All seasons were not normally distributed and Levene's test indicated unequal variances, the Kruskal-Wallis test was therefore used to test seasonal differences in ``overall_aqi``. To identify which seasonal pairs differed, pairwise Mann-Whitney U tests with Bonferroni correction were implemented.

To test whether the frequency of ``poor_air_quality`` days differed by season, a chi-square test of independence was used.

Differences for weekday/weekend were tested as well, and because weekday and weekend ``overall_aqi`` did not have normal distributions, in order to compare ``overall_aqi`` by weekday/weekend status the Mann-Whitney U test was used. Also, to test whether the frequency of ``poor_air_quality`` days showed any differences between weekdays and weekends, a chi-square test was used.

Model Selection and Justification

There were four candidate classification models chosen for this study: Logistic Regression, Decision Tree, Random Forest, and SVM.

In order to have an interpretable baseline model for binary classification, Logistic Regression was included. A model capable of nonlinear decision rules whilst remaining relatively interpretable made Decision Tree the best next candidate, whereas Random Forest can capture nonlinear relationships, reduce overfitting relative to a single tree, and provide feature importance rules. Lastly, for a nonlinear margin based classifier that could be used for comparison, SVM was chosen.

The dataset for this study contained 731 daily observations, therefore, no deep learning was used as the dataset was too small for deep learning methods. Feasibility and interpretability was prioritized for this study and therefore informed the model selection.

Data Splitting and Validation

A stratified train-test split of 80/20 was used for the main model which preserved the target class distribution in both training and test sets. The training set contained 584 observations, with 464 non-poor air quality days and 120 poor air quality days. The test set contained 147 observations, with 117 non-poor air quality days and 30 poor air quality days.

To evaluate model stability across multiple folds, stratified 5-fold cross-validation was used, and in response to moderator feedback, a chronological validation check was added. The first 80% of dates were used for training, and the final 20% of dates were used for testing during chronological validation. As a result, a more realistic time ordered validation check for environmental monitoring was provided.

Class Imbalance Handling

The target was moderately imbalanced with the final target distribution containing 581 non-poor air quality days and 150 poor air quality days. In order to reduce bias toward the majority class, class weighting was implemented for applicable baseline models.

As an additional class imbalance strategy, Synthetic Minority Oversampling Technique (SMOTE) was tested, and only applied to the training data after the train-test split. To avoid data leakage and preserve unbiased evaluation on real observations, the test set was left unchanged. Baseline class-weighted models were compared against SMOTE enhanced models.

Model Evaluation Metrics

The classification models were evaluated using accuracy, precision, recall, F1-score, ROC-AUC, classification reports, and confusion matrices.

Table 6

Model Evaluation Metrics

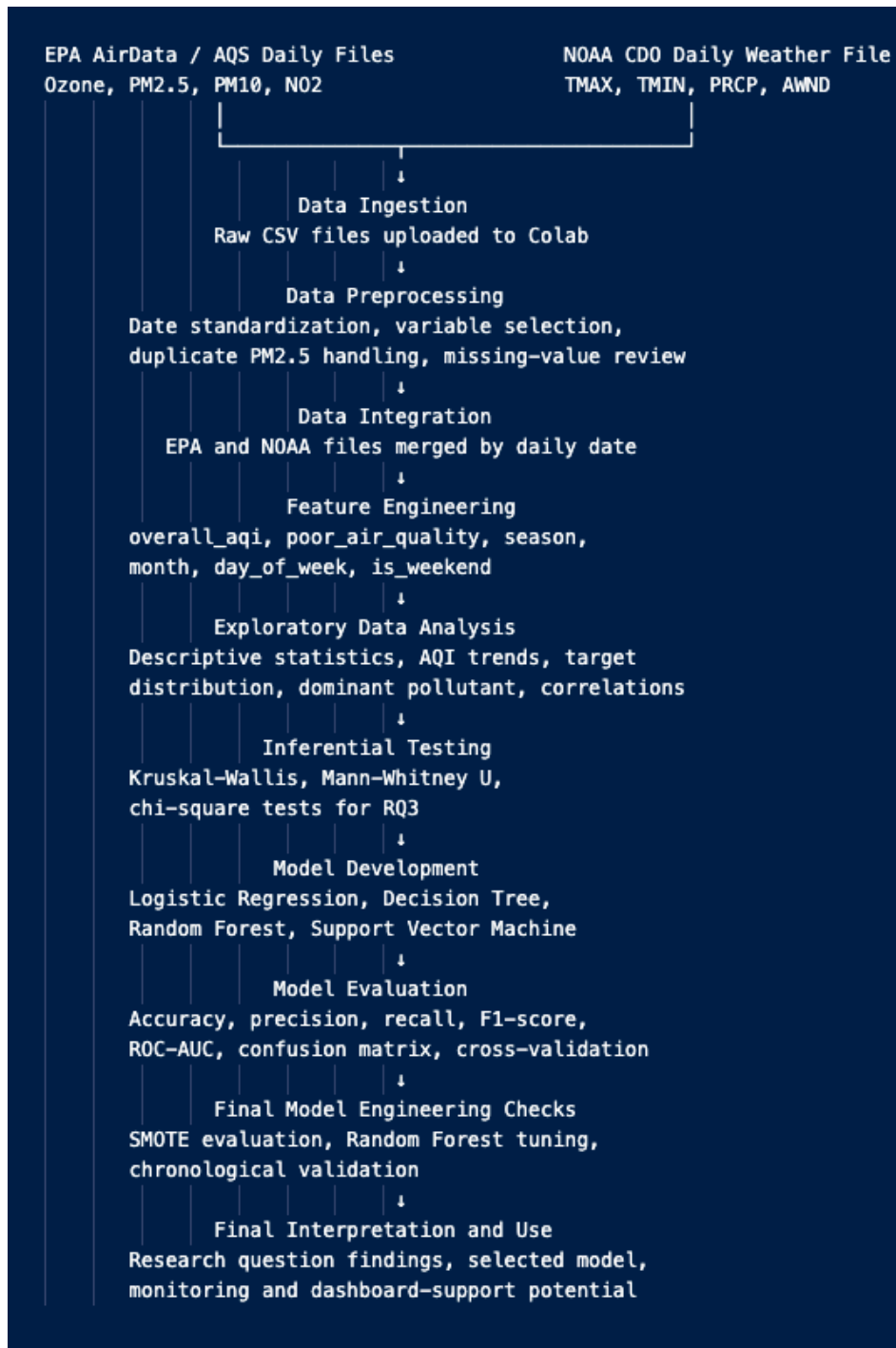
Metric	Formula	Interpretation
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall proportion of correct classifications
Precision	$TP / (TP + FP)$	Proportion of predicted poor air quality days that were truly poor air quality days

Recall	$TP / (TP + FN)$	Proportion of actual poor air quality days correctly identified
F1-score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Balance between precision and recall
ROC-AUC	Area under the ROC curve	Ability to distinguish between classes across thresholds

Architecture diagram/Workflow

System Overview

The project workflow begins with raw public data sources and proceeds through data cleaning, exploratory data analysis, model development, and result interpretation. Each stage is documented in the GitHub repository with raw and processed data files, notebooks, figures, and output tables to ensure reproducibility. This approach meets the capstone requirement for accessible data and code during evaluation.

Figure 1*Architecture Diagram for Air Quality Risk Prediction*

Workflow Components

Data Ingestion

Raw CSV files from EPA AirData / AQS and NOAA CDO Daily Summaries were used for data ingestion. EPA files provided daily pollutant records for ozone, PM2.5, PM10, and nitrogen dioxide from the Kapolei monitoring site. NOAA files supplied daily weather records from the Honolulu International Airport station. All files were uploaded to Google Colab and imported into Python with Pandas.

Data Preprocessing

Preprocessing standardized date fields across EPA and NOAA files, selected relevant variables, and reviewed missing values. Each EPA pollutant file was cleaned using a consistent function-based method to address differences in pollutant-specific variables. Duplicate PM2.5 records from multiple parameter occurrence codes were aggregated to ensure one daily observation per date.

After merging, missing numeric predictor values were reviewed and imputed using the median. Scaling was applied only for models sensitive to feature scale, such as Logistic Regression and Support Vector Machine.

Exploratory Data Analysis

Exploratory data analysis included summary statistics, missing-value review, target distribution, pollutant-specific AQI trends, seasonal and monthly summaries, dominant

pollutant analysis, and correlation analysis. These findings informed the threshold for `poor\_air\_quality` and clarified pollutant and seasonal patterns.

Feature Engineering

Feature engineering created `overall\_aqi`, `poor\_air\_quality`, `year`, `month`, `day\_of\_week`, `is\_weekend`, and season. `Overall\_aqi` was defined as the maximum pollutant-specific AQI for each date. `Poor\_air\_quality`, the binary target, was set using the threshold `overall\_aqi > 35`.

Model Development

Model development compared four supervised classifiers: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine. The feature set included pollutant concentrations, meteorological variables, and selected time-based features. Pollutant-specific AQI values and `overall\_aqi` were excluded from predictors since they defined the target.

Model Evaluation

Model performance was assessed using accuracy, precision, recall, F1-score, ROC-AUC, classification reports, and confusion matrices. The primary comparison used an 80/20 stratified train-test split, with stratified 5-fold cross-validation for stability. SMOTE was applied to training data to test synthetic oversampling for minority-class improvement. Random Forest underwent hyperparameter tuning, and chronological validation was used as a time-based sensitivity check.

Deployment

The model was not deployed as a live system for this capstone. The workflow, however, is structured to support a future dashboard or decision-support tool. A future implementation could accept daily pollutant and weather inputs, apply the classification model, and indicate whether a day is classified as locally elevated poor air quality. Further validation would be required before operational use.

Tools and Technology

The following tools and technologies were used in the project:

Table 7

Tools and Technologies

Tool / Technology	Purpose
Python	Main programming language for analysis and modeling
Google Colab	Cloud notebook environment for coding and execution
Pandas	Data loading, cleaning, merging, and table creation
NumPy	Numerical operations

Matplotlib	Data visualization and figure generation
SciPy	Inferential statistical testing
Scikit-learn	Machine learning models, preprocessing, train-test split, cross-validation, metrics, and hyperparameter tuning
Imbalanced-learn	SMOTE class-imbalance evaluation
GitHub	Repository for data files, notebooks, figures, tables, and documentation

Results

Overview of Final Analytical Dataset

The final cleaned dataset included 731 daily observations from January 1, 2024 to December 31, 2025. Each row corresponded to a calendar date for Kapolei, Honolulu County, Hawaii. The binary target, `'poor_air_quality'`, was derived from `'overall_aqi'` using the threshold `'overall_aqi > 35'`.

Table 8*Final Target Distribution*

Target Class	Description	Number of Days	Percentage
0	Not poor air quality	581	79.48%
1	Poor air quality	150	20.52%
Total		731	100.00%

The target distribution was moderately imbalanced but provided sufficient positive cases for binary classification. The chosen threshold resulted in 150 positive cases, which was more suitable for model training and evaluation than the original threshold of ``overall_aqi > 50``, which yielded only 12 positives.

Exploratory Data Analysis Results

Exploratory analysis showed ``overall_aqi`` ranged from 12 to 56 during the study period. The mean was 29.40 and the median was 29.00, indicating most days were low to moderate in AQI. This supports using a study-specific local threshold instead of an official high-AQI threshold.

Table 9*Seasonal Summary of Overall AQI and Poor Air Quality Days*

Season	Days	Mean Overall AQI	Median Overall AQI	Standard Deviation	Minimum	Maximum	Poor Air Quality Days
Winter	181	33.188	33	7.887	14	55	69
Spring	184	32.636	32	6.308	19	56	48
Summer	184	24.935	23	6.995	12	52	13
Fall	182	26.89	26	6.656	13	49	20

Winter and spring had the highest mean `overall\_aqi` values and the most poor air quality days. Summer had the lowest mean `overall\_aqi` and the fewest poor air quality days.

Table 10*Dominant Pollutant Counts*

Dominant Pollutant	Number of Days
Ozone	474
PM2.5	256

PM10	1
------	---

Ozone was the most frequent dominant pollutant, followed by PM2.5. These were the main daily AQI contributors in the Kapolei dataset.

Inferential Testing Results for RQ3

RQ3 examined whether poor air quality days varied significantly across seasons and time periods. Inferential testing was conducted at the 5% significance level.

Table 11

Final Inferential Test Results for RQ3

Analysis Area	Outcome	Grouping Variable	Primary Test	Test Statistic	df	p-value	Decision
Seasonal differences in overall AQI	overall_aqi	Season	Kruskal-Wallis	171.0418	3	< .001	Reject H ₀
Seasonal differences in poor	poor_air_quality	Season	Chi-square test of	68.4407	3	< .001	Reject H ₀

air quality frequency			independence				
Weekday vs. weekend difference s in overall AQI	overall_aqi	is_weekend	Mann-Whitney U	51,982.50	—	0.349	Fail to reject H_0
Weekday vs. weekend difference s in poor air quality frequency	poor_air_quality	is_weekend	Chi-square test of independence	0.0276	1	0.868	Fail to reject H_0

The Kruskal-Wallis test showed that `overall\_aqi` differed significantly across seasons, $H = 171.04$, $p < .001$. Pairwise Mann-Whitney U tests with Bonferroni correction showed that winter and spring were not significantly different from each other, but most other seasonal pairs differed significantly. A chi-square test also showed that the frequency of poor

air quality days differed significantly by season, $\chi^2(3) = 68.44$, $p < .001$. These results support the conclusion that poor air quality conditions varied significantly by season.

The weekday/weekend results were not statistically significant. The Mann-Whitney U test showed no significant difference in 'overall_aqi' between weekdays and weekends, $U = 51,982.50$, $p = .349$. The chi-square test also showed no significant association between weekday/weekend status and poor air quality frequency, $\chi^2(1) = 0.03$, $p = .868$. Therefore, the time-period variation observed in this study was seasonal rather than weekday/weekend based.

Table 12

Poor Air Quality Days by Season

Season	Not Poor Air Quality	Poor Air Quality	Total Days	Poor Air Quality Percentage
Winter	112	69	181	38.12%
Spring	136	48	184	26.09%
Summer	171	13	184	7.07%
Fall	162	20	182	10.99%

Winter had the highest proportion of poor air quality days at 38.12%, with spring at 26.09%. Summer and fall had substantially lower proportions.

Model Performance

Four candidate models were evaluated for binary classification: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine. The primary evaluation used an 80/20 stratified train-test split. Baseline class-weighted models were compared with SMOTE-enhanced models. Hyperparameter tuning and chronological validation were also performed for Random Forest.

Table 13

Final Model Performance Comparison: Baseline and SMOTE Models

Model	Approach	Accuracy	Precision	Recall	F1-score	ROC-AUC	Key Observation
Random Forest	Baseline class-weighted	0.986	0.967	0.967	0.967	0.994	Best overall model
Decision Tree	Baseline class-weighted	0.973	0.906	0.967	0.935	0.992	Strong second-best model

Random Forest	SMOTE training only	0.959	0.853	0.967	0.906	0.992	Same recall as baseline RF but lower precision
Decision Tree	SMOTE training only	0.952	0.829	0.967	0.892	0.988	Lower F1-score than baseline Decision Tree
Support Vector Machine	Baseline class-weighted	0.918	0.737	0.933	0.824	0.976	High recall but lower precision

Support Vector Machine	SMOTE training only	0.912	0.73	0.9	0.806	0.959	SMOTE did not improve SVM
Logistic Regression	SMOTE training only	0.891	0.694	0.833	0.758	0.935	Slight improve ment over baseline Logistic Regression
Logistic Regression	Baseline class-weighted	0.878	0.658	0.833	0.735	0.934	Interpret able baseline but weakest overall

The baseline class-weighted Random Forest model performed best overall, achieving accuracy of 0.986, precision of 0.967, recall of 0.967, F1-score of 0.967, and ROC-AUC of

0.994. The model balanced false positives and false negatives effectively on the holdout test set.

SMOTE Evaluation

SMOTE was tested as a class-imbalance strategy by applying synthetic minority oversampling to the training data only. The original test set remained unchanged to preserve unbiased evaluation on real observations.

Table 14

Effect of SMOTE on Model Performance

Model	Baseline	SMOTE	F1	Baseline	SMOTE	Precision	Baseline	SMOTE	Recall
	F1-score	F1-score	Change	Precision	Precision	Change	Recall	Recall	Change
Logistic Regression	0.735	0.758	0.022	0.658	0.694	0.037	0.833	0.833	0

Support Vector Machine	0.824	0.806	-0.018	0.737	0.73	-0.007	0.933	0.9	-0.033
Decision Tree	0.935	0.892	-0.043	0.906	0.829	-0.078	0.967	0.967	0
Random Forest	0.967	0.906	-0.06	0.967	0.853	-0.114	0.967	0.967	0

SMOTE did not improve the best-performing model. For Random Forest, SMOTE maintained recall at 0.967 but reduced precision from 0.967 to 0.853, lowering the F1-score from 0.967 to 0.906. The SMOTE Random Forest identified the same number of actual poor air quality days but produced more false positives. The baseline class-weighted Random Forest was selected as the final model.

Hyperparameter Tuning and Chronological Validation

Hyperparameter tuning for Random Forest used randomized search with stratified 5-fold cross-validation and F1-score as the optimization metric. The best parameters were 300 trees, `max\_depth` of 10, `min\_samples\_leaf` of 2, `min\_samples\_split` of 2, and `max\_features` set to None.

Table 15*Random Forest Variant Comparison*

Random Forest Version	Accuracy	Precision	Recall	F1-score	ROC-AUC	Key Observation
Baseline class-weighted	0.986	0.967	0.967	0.967	0.994	Selected final model
Tuned class-weighted	0.986	0.967	0.967	0.967	0.993	Same F1-score, slightly lower

						RO C-A UC
SMOTE training only	0.959	0.853	0.967	0.906	0.992	Low er prec isio n and F1-s core

Tuning did not improve the selected model. The tuned Random Forest matched the baseline on accuracy, precision, recall, and F1-score, but had a slightly lower ROC-AUC. The baseline class-weighted Random Forest remained the final model.

Chronological validation addressed the time-ordered nature of environmental monitoring. The first 80% of dates were used for training and the final 20% for testing.

Table 16

Chronological Validation Result for Random Forest

Model	Training Period	Test Period	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest	2024-01-01 to 2025-08-06	2025-08-07 to 2025-12-31	0.993	1	0.9	0.947	0.996

Chronological validation supported the stability of the Random Forest model under a time-ordered split. The chronological test set included only 10 poor air quality days, so results should be interpreted with caution.

Final Selected Model

The final selected model was the baseline class-weighted Random Forest. The confusion matrix for the holdout test set is shown in Table 13.

Table 17*Final Random Confusion Matrix*

Column 1	Predicted Not Poor Air Quality	
	Predicted Poor Air Quality	
Actual Not Poor Air Quality	116	1
Actual Poor Air Quality	1	29

The model correctly classified 116 non-poor air quality days and 29 poor air quality days, with one false positive and one false negative. This demonstrates a strong balance between detecting poor air quality days and limiting false alarms.

Feature Importance

Random Forest feature importance identified which predictors contributed most to classification performance. These values reflect model-based predictive contribution, not causal relationships.

Table 18*Final Random Forest Feature Importance*

Feature	Importance
daily_max_8hr_ozone	0.327
daily_mean_pm25	0.2693

daily_mean_pm10	0.1092
tmax	0.0912
month	0.0781
tmin	0.044
daily_max_1hr_no2	0.0372
awnd	0.0292
prcp	0.0122
is_weekend	0.0026

Ozone and PM2.5 were the strongest model-based predictors. PM10, maximum temperature, and month contributed at lower levels. The low importance of `is\_weekend` aligns with inferential test results showing no significant weekday or weekend effect.

Visual Evidence

Figure 2

Overall AQI by Season

The figure demonstrates that overall AQI values were elevated during winter and spring compared to summer and fall. This observed seasonal pattern aligns with the significant Kruskal-Wallis test result, confirming that `overall\_aqi` varies by season.

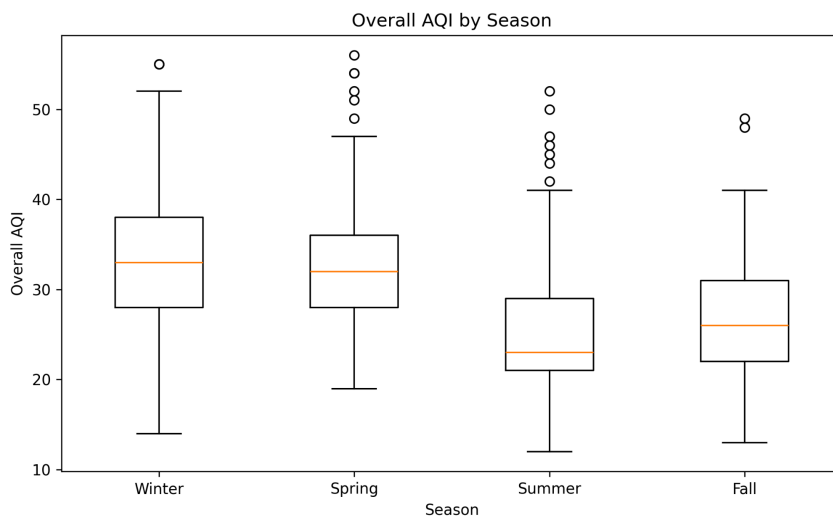


Figure 3

Final Model Comparison by F1-score

The figure presents a side-by-side comparison of the baseline, SMOTE-applied, and hyperparameter-tuned models. The baseline class-weighted Random Forest model produced the highest F1-score among all candidates.

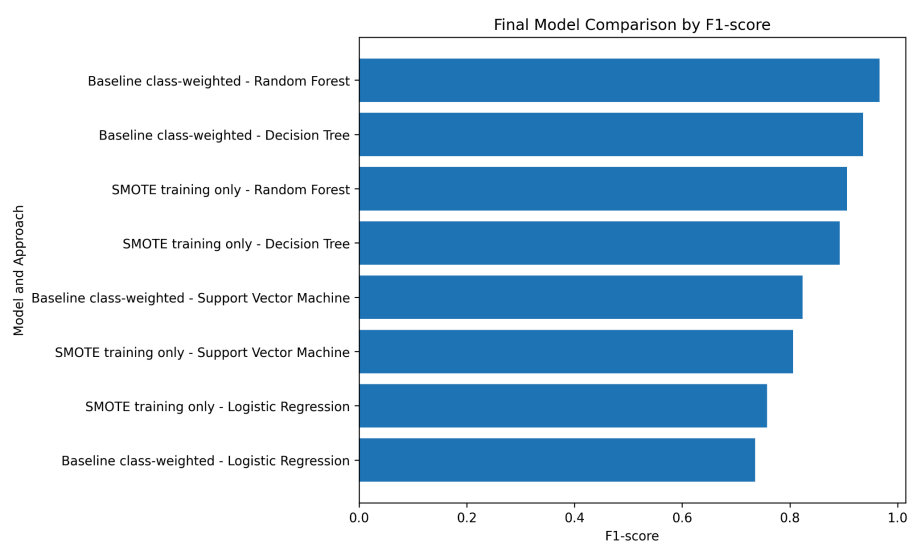


Figure 4*Final Random Forest Confusion Matrix*

The figure indicates that the final Random Forest model accurately classified almost all test set observations, resulting in a single false positive and a single false negative.

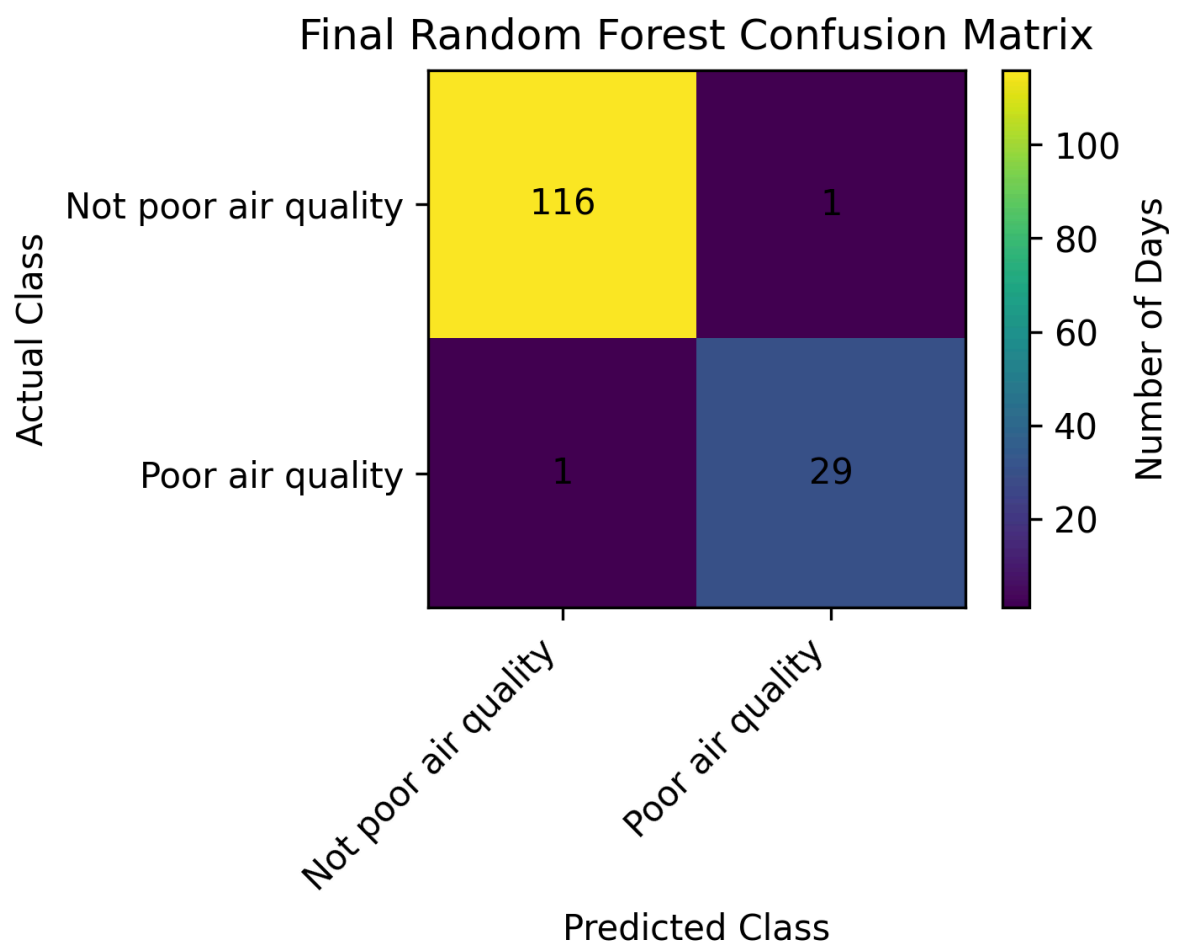
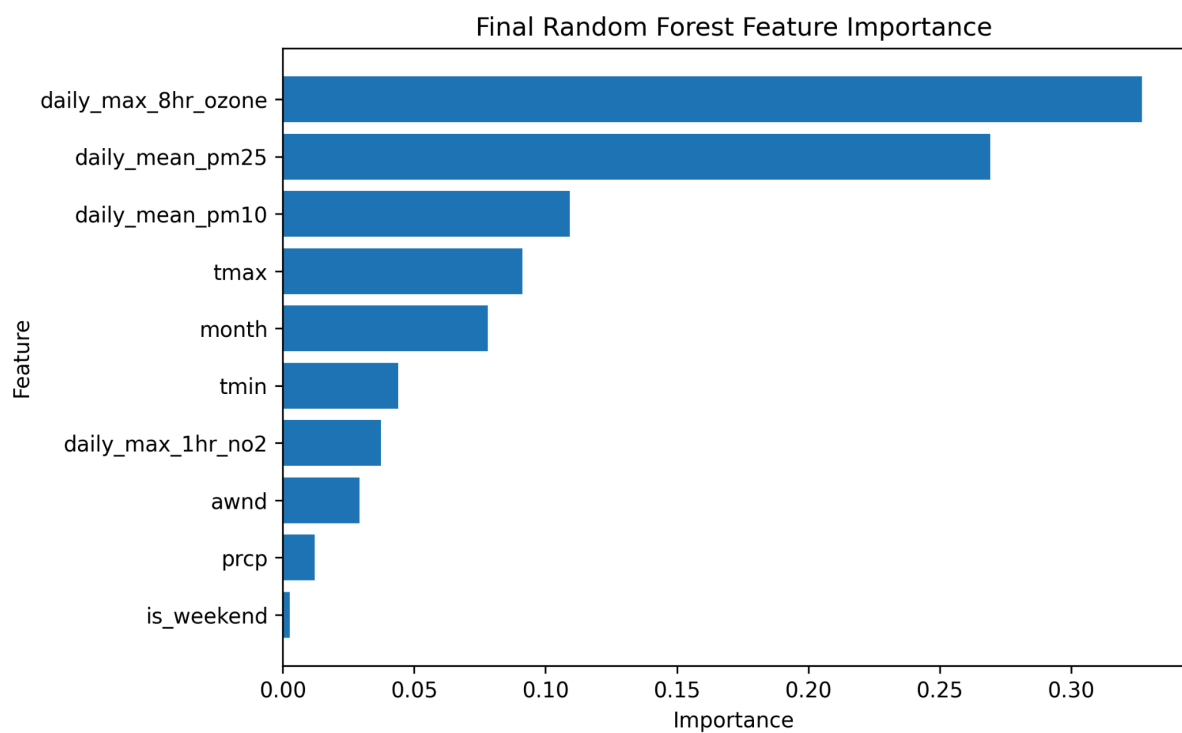


Figure 5*Final Random Forest Feature Importance*

The figure identifies ozone and PM2.5 as the most influential predictors in the final Random Forest model.

**Results by Research Question**

RQ1: Which environmental and meteorological variables are most strongly associated with poor air quality days?

The strongest model-based predictors were `daily\_max\_8hr\_ozone` and `daily\_mean\_pm25`, followed by `daily\_mean\_pm10`, `tmax`, and `month`. These results

confirm that ozone and PM2.5 were the most important variables for classifying locally elevated poor air quality days in this dataset.

RQ2: Can poor air quality days be accurately identified and classified using pollutant and weather data?

Yes. The final baseline class-weighted Random Forest model achieved F1-score = 0.967 and ROC-AUC = 0.994 on the holdout test set. The confusion matrix showed 29 true positives, 116 true negatives, one false positive, and one false negative.

RQ3: Do poor air quality days vary significantly across seasons and time periods?

Yes for season, but not for weekday/weekend status. The Kruskal-Wallis test showed that `overall\_aqi` differed significantly by season, $H = 171.04$, $p < .001$. The chi-square test showed that poor air quality frequency differed significantly by season, $\chi^2(3) = 68.44$, $p < .001$. Weekday/weekend tests were not significant.

RQ4: Which predictive model performs best for the binary classification of poor air quality?

The baseline class-weighted Random Forest model performed best, outperforming Logistic Regression, Decision Tree, Support Vector Machine, and SMOTE-enhanced variants based on F1-score and ROC-AUC.

Overall Interpretation

The results show that same-day binary classification of locally elevated poor air quality days is feasible using pollutant concentrations, meteorological variables, and

time-based features. Random Forest performed best due to its ability to capture nonlinear patterns and interactions while remaining interpretable through feature importance.

The strong performance should be interpreted with caution. While `overall\_aqi` and pollutant-specific AQI values were excluded from predictors, the target was derived from pollutant-specific AQI values and the model used pollutant concentration predictors. The model therefore learned from variables closely related to the target. These results support classification performance within the constructed dataset but do not demonstrate a fully operational forecasting system.

Practical Significance

The results show that a structured data analytics workflow can translate daily pollutant, weather, and time-based variables into a simple classification output. This approach may help environmental monitoring teams and sustainability planners identify days with locally elevated air quality and understand which variables drive the classification. The model is not intended to replace official air quality monitoring systems but can support local monitoring, planning, and future dashboard-based decision support.

Implementation and User Benefit

Deployment Approach

This implementation is designed as a local environmental monitoring and decision-support workflow, not as a full production system. The selected class-weighted Random Forest model can be integrated into a dashboard or reporting tool that processes

daily pollutant, meteorological, and temporal inputs to deliver a same-day classification indicating whether local air quality is elevated.

Deployment can be achieved using Python-based tools such as Streamlit, Flask, or Dash. The backend loads the trained model, applies the established preprocessing steps, and generates a binary output for `'poor_air_quality'`. The frontend displays the classification result, key pollutant and weather values, recent AQI trends, and model interpretation outputs such as feature importance. This system does not replace official EPA Air Quality Index reporting. It is intended as a supplemental decision-support tool for local monitoring and planning.

System Integration

Integration into an environmental monitoring workflow involves connecting daily EPA pollutant data and NOAA meteorological data to the same cleaning and feature engineering process used in this study. The system requires daily data ingestion, data validation, missing value handling, feature creation, model scoring, and results display.

Periodic retraining is required as new daily observations become available, since pollutant behavior, weather patterns, and seasonal conditions can shift over time. Before operational use, the model must be tested on additional future data and reviewed by environmental stakeholders to confirm that the study-specific threshold remains suitable for decision-support.

User Interaction

Users interact with the system through a dashboard or report interface. By uploading or connecting daily pollutant and weather data, the system returns a classification result:

- 0 = Not poor air quality
- 1 = Poor air quality

The dashboard displays the main contributing variables for interpretation. If a day is classified as poor air quality, the dashboard shows ozone, PM2.5, PM10, temperature, and month values to clarify the conditions linked to the classification. This approach provides more transparency than a black-box model output.

Benefits to Users

Primary users include environmental monitoring teams, local government planners, sustainability teams, and public information units. The model consolidates multiple daily environmental and meteorological measures into a single classification outcome, reducing the need to manually review separate pollutant and weather variables.

Environmental monitoring teams can use the framework to identify days requiring closer review. Sustainability teams can apply the output to local environmental reporting and planning. Public information units can use the classification to communicate local air quality conditions more clearly. Planners can use seasonal findings to target periods when increased monitoring is needed.

The project's strategic value lies in its reproducible workflow. By using publicly available EPA and NOAA data, open-source Python tools, and GitHub documentation, the

workflow can be reviewed, repeated, and extended. This supports transparency and simplifies adaptation for future monitoring or dashboard development.

Example Use Case

A local sustainability team reviews daily air quality conditions in Kapolei during a period of elevated AQI values. The team loads daily pollutant and weather data into the dashboard. The system applies the established preprocessing steps, calculates the required predictors, and runs the Random Forest classification model. The output classifies the day as either not poor air quality or poor air quality based on the study-specific threshold.

If the day is classified as poor air quality, the dashboard displays the relevant pollutant and weather inputs and highlights the most important model features. The team reviews whether ozone, PM_{2.5}, temperature, or seasonal timing contributed to the classification. This information supports internal monitoring, planning decisions, and future reporting. The result is not an official public-health warning, but an analytical decision-support output that helps users interpret daily local air quality conditions efficiently.

Limitations and Further Improvements

Limitations

This study is subject to several limitations. The target variable, `poor\_air\_quality`, was defined using a study-specific threshold of `overall\_aqi > 35`. This threshold was chosen based on the local AQI distribution and class balance, increasing the number of poor air quality days from 12 to 150 compared to the original $\text{overall_aqi} > 50$ cutoff. While this

approach improved binary classification feasibility, it does not align with official Environmental Protection Agency health-category thresholds. As a result, the outcome represents a local operational definition of elevated AQI, not an official unhealthy air quality category.

The study is geographically and temporally restricted to Kapolei, Honolulu County, Hawaii, from January 1, 2024 to December 31, 2025. This narrow scope supports the capstone objectives but limits generalizability. The model cannot be assumed to apply to other locations, climates, or time periods without additional validation.

Meteorological data was sourced from the NOAA Honolulu International Airport weather station, which provided daily values for maximum temperature, minimum temperature, precipitation, and average wind speed. This station is not co-located with the EPA Kapolei monitoring site. Local weather differences between the two sites may affect the accuracy of meteorological analysis.

Missing numeric predictor values were addressed using median imputation, which preserved all 731 daily observations and allowed for consistent model comparison. However, this method reduces natural variability and may not fully capture actual pollutant or weather conditions on days with missing data.

The model functions as a same-day classification tool, not a forecasting model. It relies on same-day pollutant concentrations, meteorological, and temporal variables to classify poor air quality days. It does not incorporate lagged variables, rolling averages, or future weather forecasts for prediction.

A methodological caution concerns the relationship between predictors and the target. The model excluded `overall\_aqi` and pollutant-specific AQI variables to prevent direct target leakage. However, since the target is derived from pollutant-specific AQI values and the model uses pollutant concentration variables closely tied to AQI calculation, this relationship may partly account for strong model performance and limits claims about forecasting capability.

Impact of Limitations

These limitations shape the interpretation of results. The study-specific threshold confines findings to identifying locally elevated AQI days in the Kapolei dataset, not official public-health alerts. The location-specific design restricts generalizability, and the weather-station mismatch may reduce meteorological precision. Median imputation may also smooth predictor variation, particularly on days with missing data.

The same-day classification design is useful for understanding and classifying observed conditions but does not offer early-warning forecasting. High Random Forest performance indicates that the selected variables classify the constructed target effectively within this dataset. This result should not be interpreted as evidence that the model can forecast future poor air quality without further validation and forecasting-specific features. The classification framework could be extended into a true forecasting model by adding lagged predictors such as previous-day pollutant concentrations, previous-day `overall\_aqi`, rolling pollutant averages, and multi-day

weather summaries. This would allow the model to predict future air quality conditions rather than classify same-day observations.

Future work should test additional validation strategies using a longer time series. While this study included chronological validation, the test set contained only 10 poor air quality days. Expanding the dataset would support more robust time-based validation and assess model stability across years.

Future research should include additional local monitoring stations or compare multiple locations across Hawaii. This would test the generalizability of predictors and model structure beyond Kapolei and help identify differences in seasonal patterns across coastal, urban, or inland areas.

Future model development should test additional feature-engineering approaches, such as humidity, atmospheric pressure, wind direction, interaction terms, rolling averages, or pollutant ratios, if reliable data are available. These features may improve interpretability and model performance, particularly if the project expands to forecasting.

Future tuning should extend beyond Random Forest. While this study focused on Random Forest hyperparameter tuning due to its strong performance, future work should also tune Support Vector Machine, compare gradient boosting methods, and test calibrated probability models if probability-based thresholds are needed for implementation.

Future Scope

The future scope of this project includes developing a local environmental monitoring dashboard or decision-support tool. Such a system would automatically ingest daily pollutant

and meteorological data, apply the trained model, and display classification outputs with supporting visualizations. Users would be able to review daily classifications of locally elevated poor air quality and identify the variables contributing most to each result.

The project can also support sustainability planning and environmental reporting. Seasonal analysis indicates that winter and spring require closer monitoring in the Kapolei dataset. A future dashboard should display seasonal trends, pollutant contributions, and model classifications to inform planning and public information efforts.

Before operational deployment, the model requires further validation on future data, review by environmental stakeholders, and comparison with official AQI reporting practices. The study-specific threshold must also be reviewed with domain experts before use in public communication.

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Appendix A: Data Dictionary

Table A1

Data Dictionary for the Final Cleaned Analytical Dataset

Variable Name	Description	Source	Data Type	Unit / Values	Role in Study
date	Calendar date for the daily observation	EPA / NOAA	Date	YYYY-MM-DD	Merge key / identifier
site_id	Unique EPA monitoring site identifier	EPA	String / Integer	Not applicable	Site metadata
local_site_name	Name of the EPA monitoring site	EPA	String	Kapolei	Site metadata
county	County where the EPA	EPA	String	Honolulu	Site metadata

	monitoring site is located				
state	State where the EPA monitoring site is located	EPA	String	Hawaii	Site metadata
site_latitude	Latitude of the EPA monitoring site	EPA	Float	Decimal degrees	Site metadata
site_longitude	Longitude of the EPA monitoring site	EPA	Float	Decimal degrees	Site metadata
station	NOAA weather station identifier	NOAA	String	Station code	Weather station metadata

station_name	NOAA weather station name	NOAA	String	Honolulu International Airport, HI US	Weather station metadata
daily_max_8 hr_ozone	Daily maximum 8-hour ozone concentration	EPA	Float	ppm	Predictor
daily_mean_ pm10	Daily mean PM10 concentration	EPA	Float	µg/m³	Predictor
daily_mean_ pm25	Daily mean PM2.5 concentration	EPA	Float	µg/m³	Predictor
daily_max_1 hr_no2	Daily maximum 1-hour nitrogen	EPA	Float	ppb	Predictor

	dioxide concentratio n				
daily_aqi_oz one	Daily AQI value associated with ozone	EPA	Float / Integer	AQI index	Target-constr uction variable
daily_aqi_p m10	Daily AQI value associated with PM10	EPA	Float / Integer	AQI index	Target-constr uction variable
daily_aqi_p m25	Daily AQI value associated with PM2.5	EPA	Float / Integer	AQI index	Target-constr uction variable
daily_aqi_no 2	Daily AQI value associated with nitrogen dioxide	EPA	Float / Integer	AQI index	Target-constr uction variable

daily_obs_count_ozone	Number of daily observations used for ozone summary	EPA	Float / Integer	Count	Data quality variable
daily_obs_count_pm10	Number of daily observations used for PM10 summary	EPA	Float / Integer	Count	Data quality variable
daily_obs_count_pm25	Number of daily observations used for PM2.5 summary	EPA	Float / Integer	Count	Data quality variable
daily_obs_count_no2	Number of daily observations	EPA	Float / Integer	Count	Data quality variable

	used for nitrogen dioxide summary				
percent_com plete_ozone	Percentage completeness of ozone observations for the date	EPA	Float	Percent	Data quality variable
percent_com plete_pm10	Percentage completeness of PM10 observations for the date	EPA	Float	Percent	Data quality variable
percent_com plete_pm25	Percentage completeness of PM2.5 observations for the date	EPA	Float	Percent	Data quality variable
percent_com plete_no2	Percentage completeness	EPA	Float	Percent	Data quality variable

	s of nitrogen dioxide observations for the date				
ozone_poc	Parameter occurrence code value(s) for ozone records	EPA	String / Float	POC code	Data quality / tracking variable
pm10_poc	Parameter occurrence code value(s) for PM10 records	EPA	String / Float	POC code	Data quality / tracking variable
pm25_poc	Parameter occurrence code value(s) for	EPA	String / Float	POC code	Data quality / tracking variable

	PM2.5 records				
no2_poc	Parameter occurrence code value(s) for nitrogen dioxide records	EPA	String / Float	POC code	Data quality / tracking variable
awnd	Average daily wind speed	NOAA	Float	meters per second	Predictor
prcp	Daily precipitation	NOAA	Float	millimeters	Predictor
tmax	Daily maximum temperature	NOAA	Float	degrees Celsius	Predictor
tmin	Daily minimum temperature	NOAA	Float	degrees Celsius	Predictor

overall_aqi	Daily overall AQI calculated as the maximum available pollutant-specific AQI value	Derived	Float / Integer	AQI index	Outcome-construction variable
poor_air_quality	Binary target indicating whether the day is classified as locally elevated poor air quality	Derived	Integer	0 = Not poor, 1 = Poor	Target variable
year	Year extracted from the date	Derived	Integer	2024 or 2025	Time feature

month	Month extracted from the date	Derived	Integer	1–12	Predictor / time feature
day_of_week	Day of week extracted from the date	Derived	String	Monday–Sunday	Time feature
is_weekend	Indicator showing whether the date occurred on a Saturday or Sunday	Derived	Integer	0 = Weekday, 1 = Weekend	Predictor / time feature
season	Calendar season assigned from the month	Derived	String	Winter, Spring, Summer, Fall	Time feature / RQ3 grouping variable

dominant_pollutant	Pollutant with the highest pollutant-specific AQI value for the date	Derived	String	Ozone, PM2.5, PM10, NO2	EDA / interpretation variable
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Note: The model predictor set included pollutant concentration variables, meteorological variables, month, and `is\_weekend`. Pollutant-specific AQI variables and `overall\_aqi` were excluded from the predictors since they were used to define the target variable.